

• Here we study fields with spin $s > 2$

• Useful fields $s \geq 0$:

$\rightarrow s = 0$: scalars

$\rightarrow s = \frac{1}{2}$: matter fields (e^- , q , etc.)

$\rightarrow s = 1$: gauge fields (photons, gluons, ...)

$\rightarrow s = \frac{3}{2}$: gravitino

$\rightarrow s = 2$: graviton

} SM

• Why study $s > 2$ fields?

$\rightarrow$ they appear in ST

$\rightarrow$ theoretical challenge

$\rightarrow$ 1987 Vasiliev's equations (in AdS, ∞ of them)

$\rightarrow$ these provide a variety of AdS/CFT examples

• Outline

1. Unitary reps of Poincaré for HS fields

2. Fronsdal equation

3. Some bosonic ST

4. Review of no-go theorems (constrain HS field interactions)

• Conventions

$\rightarrow$ dim $\mathcal{D}$, $\eta = \text{diag}(-, +, \dots, +)$

$\rightarrow$ symmetrisation : $\tau_{(\mu_1 \dots \mu_k)} = \frac{1}{k!} \sum_{\sigma \in S_k} \tau_{\mu_{\sigma(1)} \mu_{\sigma(2)} \dots \mu_{\sigma(k)}}$

$$\Rightarrow \text{they carry weight one, e.g. } \partial_\mu \xi_\nu = \frac{1}{2}(\partial_\mu \xi_\nu + \partial_\nu \xi_\mu)$$

$$\rightarrow \text{light-cone coord. } x^\pm = \frac{1}{\sqrt{2}}(x^{D-1} \pm x^0) \quad x^i, i=1, \dots, D-2$$

$$\Rightarrow \eta_{\mu\nu} x^\mu x^\nu = 2x^+ x^- + \delta_{ij} x^i x^j$$

$$\Rightarrow \eta_{+-} = \eta_{-+} = 1, \quad \eta_{ij} = \delta_{ij}$$

$$\Rightarrow V^+ = V_-, \quad V_\mu W^\mu = V_+ W_- + V_- W_+ + V_i W^i$$

1. Unitary reps of Poincare

1.1. Group & algebra

$$\rightarrow x^\mu \mapsto x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu$$

$\in \mathbb{R}^D$

$\Lambda \in SO^+(D-1, 1)$

$$\det \Lambda = 1$$

$$\Lambda^0_0 \geq 1$$

$$\rightarrow (\Lambda_2, a_2) \circ (\Lambda_1, a_1) = (\Lambda_2 \Lambda_1, a_2 a_1)$$

$$\text{semi-direct prod.} \quad : \quad ISO^+(D-1, 1) \cong SO^+(D-1, 1) \ltimes \mathbb{R}^D$$

• Algebra:

$$[M_{\mu\nu}, M_{\alpha\beta}] = i(\eta_{\mu\alpha} M_{\nu\beta} + \eta_{\nu\beta} M_{\mu\alpha} - \eta_{\mu\beta} M_{\nu\alpha} - \eta_{\nu\alpha} M_{\mu\beta})$$

$$[P_\mu, P_\nu] = 0$$

$$[P_\mu, M_{\alpha\beta}] = -i(\eta_{\mu\beta} P_\alpha - \eta_{\mu\alpha} P_\beta)$$

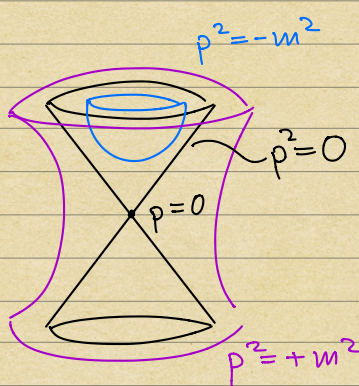
1.2 Little group

• Method of induced rep [Wigner 1939]

→ fix eigenvalues of P_μ 's (call them p_μ)

→ look at $\mathfrak{h} \subseteq SO^+(\mathcal{D}-1, 1)$ which preserves p_μ
↳ some subgroup labelled 'little group'

→ several cases:

	name	P_μ	$\mathfrak{h}$	orbits
$p^2 = -m^2$	massive	$(m, 0, \dots, 0)$	$SO(\mathcal{D}-1)$	
$p^2 = +m^2$	tachyonic	$(0, 0, \dots, m)$	$SO^+(\mathcal{D}-2, 1)$	
$p^2 = 0, p \neq 0$	massless	$(E, 0, \dots, 0, E)$	$ISO(\mathcal{D}-2)$	
$P_\mu = 0$	zero-momentum	$(0, \dots, 0)$	$SO^+(\mathcal{D}-1, 1)$	

• Claim: for $p \neq 0, p^2 = 0$ $\mathfrak{h} = ISO(\mathcal{D}-2)$

Proof: 1. light-cone coord. $(p^\mu) = (p_-, 0, \dots, 0)$

2. look carefully at $[P_\mu, M_{\nu\rho}]$ & find that $\mathfrak{h}$ is generated by $\{M_{ij}, M_{+i}\}$

3. compute the alg. of M_{ij} 's & $\pi_i = P_- M_{+i}$

⇒ you will find $ISO(\mathcal{D}-2)$

• Reps of $ISO(D-2)$ (induced reps)

→ fix eigenvalues of Π_i (labelled $\vec{\xi}_i$)

→ look at $\tilde{H} \subseteq SO(D-2)$ which fix $\vec{\xi}$ ("short little group" (little group of the little group))

	name	$\vec{\xi}$	$\tilde{H}$
$\vec{\xi}^2 = \mu^2 > 0$	as spin infinite spin	$(\mu, 0, \dots, 0)$	$SO(D-3)$
$\vec{\xi}^2 = 0$	helicity reps	$(0, \dots, 0)$	$SO(D-2)$

• Comments

→ ∞ spin: limit $s \rightarrow \infty$ of massive NS reps ($\mu = m s$ fixed)

→ we will only consider massive & helicity reps
 $SO(D-1)$ $SO(D-2)$

→ # dofs = dim of irrep of $\tilde{H}$

$\Rightarrow$ massive & helicity reps have finite # dofs

1.3. Tensorial unitary reps of Poincaré

→ these are spaces of tensors specified by

1) symm. conditions

2) tracelessness conditions

3) we only consider bosonic reps

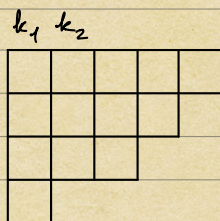
• Example: $\tau_{ij} = \frac{1}{2}(\tau_{ij} + \tau_{ji}) - \frac{1}{N} \delta_{ij} \leftarrow \text{symm. + traceless}$

$+ \frac{1}{2}(\tau_{ij} - \tau_{ji}) \leftarrow \text{a-sym.}$

$+ \frac{1}{N} \delta_{ij} \tau_{kk} \leftarrow \text{trace part}$

• Symmetries

$\hookrightarrow Y \tau$



$k_1 \geq k_2 \geq \dots \geq k_p$

decreasing heights

$\rightarrow$ tensors $\tau_{i_1 i_2 \dots i_{k_1} j_1 \dots j_{k_2} l_1 \dots l_{k_p}}$ satisfy

a) a-sym. in each group

b) a-sym. of all the indices in a group + 1 index to the right = 0